Medicine

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Introduction

This document contains my notes from Swansea University's graduate-entry medicine course. The notes are accessible through GitHub:

https://github.com/jfbeda/medicine

You are encouraged to edit and contribute to the notes as you see fit. My contact details are available on my website:

https://beda.ca/jack

0.1 Organisation

The main body of the handbook is divided into chapters. Each chapter is stored as a separate file inside the `chapters` folder.

For example:

- `chapters/02-cardiovascular.tex`
- `chapters/03-respiratory.tex`
- `chapters/04-gastrointestinal.tex`

The chapters are imported in the required order from `main.tex`.

CHAPTER 1

Administration

This chapter contains all of the initial administration processes I needed to do, and the order in which I did them.

1.1 Learning@Wales Training

1.1.1 Registering for Learning@Wales

1. Go to `learning.nhs.wales`.
2. Select **Learning@Wales**.
3. Select the button right below the text that says

    > 'Please click the button below to register for an account if you are unable to self-register.'

4. Register, and wait to be accepted.

1.1.2 Required training in Learning@Wales

Complete the following training modules:

- Information Governance, Records Management & Cyber Security
- Fire Safety
- Infection Prevention & Control Level 1
- Health & Safety
- Treat Me Fairly
- Safeguarding People

- Violence Against Women
- Violence and Aggression
- Manual Handling
- ANTT

1.2 Canvas training

- NAC - Anti-racism Course
- NAC - Enhance your EDI Skills

1.3 Resources

- Dissectible me, 5 minute anatomy
- Moore's clinically oriented anatomy
- Gray's anatomy for students
- 'kenhub'
- 'teaching anatomy'
- RHiMME track: click here Due September 25, 5pm

CHAPTER 2

Important Links

2.1 Important links

- **Student feedback**
- **Leave of absence request**

CHAPTER 3

Anatomy

3.1 Anatomical terms

3.1.1 Anatomical planes

- Coronal (frontal) plane: divides the body into anterior and posterior halves.
- Sagittal plane: divides the body into left and right halves. The **median plane** divides the body into equal left and right halves, while a **parasagittal plane** divides the body into unequal left and right halves.
- Transverse (horizontal or axial) plane: divides the body into superior and inferior halves.

3.1.2 Anatomical positional terms

- Superior
- Inferior
- Cephalic (towards the head)[1]
- Caudal (towards the tail)[1]
- Proximal (closer to the point of attachment)
- Distal (further from the point of attachment)
- Medial
- Lateral
- Posterior/dorsal
- Anterior/ventral

[1] The terms **cephalic** and **caudal** are often used in embryology, where the head and tail of the embryo are more distinct than in adults.

3.1.3 Anatomical movement terms

- Supination: rotation of the forearm so that the palm faces anteriorly (or superiorly when the arm is outstretched)
- Pronation: rotation of the forearm so that the palm faces posteriorly (or inferiorly when the arm is outstretched)
- Inversion: rotation of the foot so that the sole faces medially
- Eversion: rotation of the foot so that the sole faces laterally
- Flexion: movement that decreases the angle between two body parts[2]
- Extension: movement that increases the angle between two body parts[2]
- Abduction: movement away from the midline of the body
- Adduction: movement towards the midline of the body
- Circumduction: circular movement of a body part, combining flexion, extension, abduction, and adduction
- Rotation: movement around a longitudinal axis, either medially (internal rotation) or laterally (external rotation)
- Elevation: movement of a body part superiorly
- Depression: movement of a body part inferiorly
- Protraction: movement of a body part anteriorly
- Retraction: movement of a body part posteriorly
- Opposition: movement of the thumb to touch the tips of the other fingers
- Reposition: movement of the thumb back to its anatomical position
- Dorsiflexion: movement of the foot so that the toes point superiorly
- Plantar flexion: movement of the foot so that the toes point inferiorly
- Palmar flexion: movement of the hand so that the palm faces inferiorly

[2] the ‘angle’ being referred to here is the angle in the ‘natural’ movement (e.g. at the hip it is the raising of the leg that decreases the angle on the anterior side, which is the side that is usually considered when describing flexion and extension).

CHAPTER 4

Diarrhea

4.1 The Small Intestine

Based on the video *The small intestine: How does the small intestine absorb so much?* by Christian Cobbold.

- Small intestine has many villi and microvilli to increase surface area for absorption.
- Each epithelial cell contains order 3000 microvilli.
- Glucose is absorbed into cells via sodium-glucose transporters (SGLT1) that work by pumping sodium ions out of the cell (towards the interstitial fluid), creating a sodium gradient (with respect to the intestinal fluid), that results in sodium uptake from the intestinal fluid where glucose is co-transported with sodium into the cell.
- Fat globules are emulsified by bile salts which are then (somehow?) turned into micelles (whatever they happen to be?) which are then turned into water-soluble chylomicrons (diameter order 1 µm) that are absorbed into the lymphatic system.

4.2 The Large Intestine

Based on the video *Large intestine: Absorption and defaecation* by Christian Cobbold.

- Age results in decreased gastrointestinal motility, which can result in constipation.
- Age results in decreased large intestine muscle tone and thus straining when defecating, which can result in haemorrhoids.
- Age results in weakening sphincter muscles which results in acid reflux and heartburn.

4.3 Fluid Movement Between Body Compartments

Based on the video *Fluid movement between body compartments: Why doesn't all your fluid stay in the bloodstream?* by Christian Cobbold.

4.4 Oedema

Based on the video *Oedema: Why fluid accumulates in tissues* by Christian Cobbold.